① a) Note: (7) and (3) are given. Then, by (7), rearranging (and taking gradients w.r.p. to z, with perturbation μu, where $\mu > 0$ and $u \sim N(0, I_d)$):

$$\nabla_z L(w+\mu u, z) \approx \nabla_z L(w,z) + \left(\nabla^2_{wz} L(w,z)\right)^T (\mu u)$$

Rearrange and transpose:

$$\mu u^T \cdot \left(\nabla^2_{wz} L(w,z)\right) \approx \left(\nabla_z L(w+\mu u, z) - \nabla_z L(w,z)\right)^T$$

$$\Downarrow \cdot \frac{u}{\mu}$$

$$u u^T \left(\nabla^2_{wz} L(w,z)\right) \approx \frac{u}{\mu}\left(\nabla_z L(w+\mu u, z) - \nabla_z L(w,z)\right)^T$$

$$\Downarrow \mathbb{E}_u$$

$$\underbrace{\mathbb{E}_u[u u^T]}_{\mathbb{E}_u\begin{bmatrix} u_1^2 & \cdots \\ u_1 u_2 & \\ \vdots & u_d^2 \end{bmatrix} = \frac{1}{d} I_d,} \cdot \underbrace{\left(\nabla^2_{wz} L(w,z)\right)}_{\text{independent of } u} \approx \mathbb{E}_u\left[\frac{u}{\mu}\left(\nabla_z L(w+\mu u, z) - \nabla_z L(w,z)\right)^T\right]$$

since $u \sim N(0, I_d)$, so each $u_i \perp u_j$ if $u_i \neq u_j$, and u_i^2 is the variance (for 0-mean).

$$\Downarrow \text{consider } \mathbb{E}_u[f(u)] \approx \sum_{i=1}^{N} \frac{f(u_i)}{N} \approx f(u) \text{ for } N = 1.$$

So using that: $\mathbb{E}_u\left[\frac{u}{\mu}\left(\nabla_z L(w+\mu u, z) - \nabla_z L(w,z)\right)^T\right] \approx$

$$\approx \frac{u}{\mu}\left(\nabla_z L(w+\mu u, z) - \nabla_z L(w,z)\right)^T \text{ for some } u$$

$$\Downarrow \text{So}$$

$$\nabla^2_{wz} L(w,z) \approx \frac{u}{\mu}\left(\nabla_z L(w+\mu u, z) - \nabla_z L(w,z)\right)^T.$$ Substituting into (3):

$$\nabla_w \|\nabla_z L(w,z)\| \approx \frac{u}{\mu}\left(\nabla_z L(w+\mu u, z) - \nabla_z L(w,z)\right)^T \nabla_z L(w,z) =$$

$$= \frac{\langle \nabla_z L(w+\mu u, z), \nabla_z L(w,z)\rangle - \|\nabla_z L(w,z)\|^2}{\mu \|\nabla_z L(w,z)\|} u \quad \square$$

b) The gradient of the objective is:

$$\nabla f(w) + \rho \, \mathbb{E}_z\left[\nabla_w \|\nabla_w L(w,z)\|\right] \underset{\text{by eq(6)}}{=} \nabla f(w) + \rho \, \mathbb{E}_z\left[\nabla_w^2 L(w,z) \underbrace{\frac{\nabla_w L(w,z)}{\|\nabla L(w,z)\|}}_{u}\right] \approx$$

goes in cos $\mathbb{E}$ is over z's, so by linearity of expectation $\nabla_w \mathbb{E}_z[\cdot] = \mathbb{E}_z[\nabla_w \cdot]$

$\nabla f(w) = \mathbb{E}_z[\nabla L(w,z)]$ by definition.

$$\approx \nabla f(w) + \mathbb{E}_z\left[\nabla_w L\left(w + \rho \underbrace{\frac{\nabla_w L(w,z)}{\|\nabla_w L(w,z)\|}}_{=u}, z\right) - \nabla_w L(w,z)\right] =$$

by eq(7), and ρ goes in

$$= \mathbb{E}_z\left[\nabla_w L(w,z) + \nabla_w L(w+\rho u, z) - \nabla_w L(w,z)\right] = \mathbb{E}_z\left[\nabla_w L(w+\rho u, z)\right]$$

Finally: $\mathbb{E}_z[\nabla_w L(w+\rho u, z)] \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_w L(w+\rho u, z_i) \underset{\text{for } N=1,\, z_1=z}{=} \nabla_w L(w+\rho u, z)$.

So, combining all:

$\nabla f(w) + \rho \, \nabla \mathbb{E}_z[\nabla_w \|\nabla_w L(w,z)\|] \approx \nabla_w L(w+\rho u, z)$ for $u = \frac{\nabla_w L(w,z)}{\|\nabla_w L(w,z)\|}$, showing that $\nabla_w L(w+\rho u, z)$ is a good estimator of the gradient of eq. (2). $\square$.

(1) c) First, define $u_t = \frac{\nabla L(w_t, z_t)}{\|\nabla L(w_t, z_t)\|_2}$, note: $\|u_t\|_2 = \frac{\|\nabla L(w_t,z_t)\|_2}{\|\nabla L(w_t,z_t)\|_2} = 1$.
scalar, can take out

Then:

$$\mathbb{E}_{z_t}[\|g_t\|_2^2] \overset{\text{by definition of } g_t}{=} \mathbb{E}_{z_t}[\|\nabla L(w_t + \rho_t u_t, z_t)\|_2^2] =$$

$\|a+b\|_2^2 \le 2\|a\|_2^2 + 2\|b\|_2^2$ always true, also linearity of $\mathbb{E}$

$$= \mathbb{E}_{z_t}[\|\nabla L(w_t+\rho_t u_t, z_t) - \nabla L(w_t, z_t) + \nabla L(w_t, z_t)\|_2^2] \le$$

$$\le \mathbb{E}_{z_t}[2\|\nabla L(w_t+\rho_t u_t, z_t) - \nabla L(w_t, z_t)\|_2^2] + \mathbb{E}_{z_t}[2\|\nabla L(w_t, z_t)\|_2^2]$$

$$\le 2L^2 \mathbb{E}_{z_t}[\|\rho_t u_t\|_2^2] + 2\mathbb{E}_{z_t}\|\nabla L(w_t, z_t)\| \overset{\rho_t \text{ is a scalar, from } \|u_t\|_2^2 = 1}{=} 2L^2\rho_t^2 + 2\mathbb{E}_{z_t}\|\nabla L(w_t, z_t)\|_2^2$$

$L(\cdot, i)$ is L-smooth

$$= 2L^2\rho_t^2 + 2\|\nabla f(w_t)\|_2^2 + 2\mathbb{E}_{z_t}[\|\nabla L(w_t, z_t) - \nabla f(w_t)\|_2^2] \le$$

Since $\mathbb{E}[\nabla L(w_t, z_t)] = \nabla f(w_t)$, and $= \mathbb{E}_{z_t}[\nabla L(w_t, z_t)]$

$\mathbb{E}\|X\|_2^2 = \|\underbrace{\mathbb{E}[X]}_{\nabla f(w_t)}\|_2^2 + \mathbb{E}[\|X - \mathbb{E}[X]\|_2^2] = \|\mathbb{E}[X]\|_2^2 + \mathrm{Var}[X]$

$$\le 2L^2\rho_t^2 + 2\|\nabla f(w_t)\|_2^2 + 2\sigma^2.$$

by the assumptions on stochastic gradient $\nabla L(w_t, z_t)$

So this shows:

$$\mathbb{E}_{z_t}\|g_t\|_2^2 \le 2L^2\rho_t^2 + 2\|\nabla f(w_t)\|_2^2 + 2\sigma^2. \quad \square.$$

(2) a) First, by definition of SGM: take $\|\cdot\|^2$

$$w_{t+1} = w_t - \eta g_t \Leftrightarrow w_{t+1} - w^* = w_t - \eta g_t - w^* \Leftrightarrow$$

$$\|w_{t+1} - w^*\|^2 = \|w_t - w^* - \eta g_t\|^2 = \|w_t - w^*\| - 2\eta \langle w_t - w^*, g_t \rangle + \eta^2 \|g_t\|^2$$

$\Updownarrow$ rearrange, divide by 2η, also $\langle \cdot, \cdot \rangle$ is commutative ($\|a-b\|^2 = \|a\|^2 - 2\langle a, b \rangle + \|b\|^2$)

$$\langle g_t, w_t - w^* \rangle = \frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta} + \frac{\eta \|g_t\|^2}{2}.$$

Since g_t equals either $\partial f(w_t)$ or $\partial h(w_t)$, and f and h are G-Lipschitz, so $\|\partial f(w)\| \leq G$ and $\|\partial h(w)\| \leq G \; \forall w$, then it is always true that $\|g_t\| \leq G$ so $\|g_t\|^2 \leq G^2$. So the following holds by equivalence:

$$\langle g_t, w_t - w^* \rangle = \frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta} + \frac{\eta}{2}\|g_t\|^2 \leq \frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta} + \frac{\eta G^2}{2}.$$

$\Downarrow$ Sum over $t = 1, \dots, T$:

$$\sum_{t=1}^{T} \langle g_t, w_t - w^* \rangle \leq \frac{1}{2\eta} \left(\underbrace{\|w_1 - w^*\|^2}_{=D^2} - \|w_2 - w^*\|^2 + \|w_2 - w^*\|^2 - \|w_3 - w^*\|^2 + \dots - \|w_{T+1} - w^*\|^2 \right) + \frac{T\eta G^2}{2} =$$

$$= \frac{1}{2\eta} \left(D^2 - \underbrace{\|w_{T+1} - w^*\|^2}_{\geq 0} \right) + \frac{T\eta G^2}{2} \leq \frac{D^2}{2\eta} + \frac{T\eta G^2}{2}.$$

(telescope)

divide by T $\Leftrightarrow$

$$\frac{1}{T} \sum_{t=1}^{T} \langle g_t, w_t - w^* \rangle \leq \frac{\eta G^2}{2} + \frac{D^2}{2\eta T}. \quad \square$$

$\square$.

Showing the desired equality (it holds because $w_{t+1} = w_t - \eta g_t$ and by all assumptions).

(2) b) Since f and h are convex: $= -(w^* - w_t)$

$f(w^*) \geq f(w_t) + \langle \nabla f(w_t), w^* - w_t \rangle \Rightarrow f(w_t) - f(w^*) \leq \langle \nabla f(w_t), w_t - w^* \rangle$

$h(w^*) \geq h(w_t) + \langle \nabla h(w_t), w^* - w_t \rangle \Rightarrow h(w_t) - h(w^*) \leq \langle \nabla h(w_t), w_t - w^* \rangle$.

Now, by (9) $w^* = \underset{w}{\operatorname{argmin}} f(w)$ s.t. $h(w) \leq 0$, so by definition of this problem, $h(w^*) \leq 0$ (otherwise w^* couldnt be a solution).

So: $h(w^*) \leq 0 \Leftrightarrow -h(w^*) \geq 0 \;/ + h(w_t) \Leftrightarrow h(w_t) - h(w^*) \geq h(w_t) + 0$

So:

$$h(w_t) \leq h(w_t) - h(w^*) \underset{\text{as established by convexity first}}{\leq} \langle \nabla h(w_t), w_t - w^* \rangle.$$

Finally:

$$\frac{1}{T}\sum_{t\in A}\big(f(w_t) - f(w^*)\big) + \frac{1}{T}\sum_{t\in B} h(w_t) \underset{\text{convexity, and above}}{\leq} \frac{1}{T}\sum_{t\in A}\langle \nabla f(w_t), w_t - w^* \rangle + \frac{1}{T}\sum_{t\in B}\langle \nabla h(w_t), w_t - w^* \rangle =$$

$$= \underset{\forall t\in A,\ h(w_t)\leq \varepsilon,\ \text{so } \nabla f(w_t) = g_t\ \forall t\in A}{\underset{\forall t\in B,\ h(w_t) > \varepsilon,\ \text{so } \nabla h(w_t) = g_t\ \forall t\in B}{=}} \frac{1}{T}\left(\sum_{t\in A}\langle g_t, w_t - w^* \rangle + \sum_{t\in B}\langle g_t, w_t - w^* \rangle\right) \underset{\{t\in A\}\cup\{t\in B\} = [T]}{=} \frac{1}{T}\sum_{t=1}^{T}\langle g_t, w_t - w^* \rangle \underset{\text{by a)}}{\leq} \frac{\eta G^2}{2} + \frac{D^2}{2\eta T}$$

□

c) Since h is convex, $\forall d_i \geq 0$ s.t. $\sum_i d_i = 1$, $h\left(\sum_i d_i w_i\right) \leq \sum_i d_i h(w_i)$. Let $d_i = \frac{1}{|A|}$, $w_i = w_t$ s.t. $t \in A$ for different $t \in A$. Then:

$$h(\bar{w}) = h\left(\frac{1}{|A|}\sum_{t\in A} w_t\right) \underset{\text{by convexity of } h}{\leq} \frac{1}{|A|}\sum_{t\in A} h(w_t) \underset{\text{Since } t\in A \text{ and } A = \{t\in[T] \mid h(w_t)\leq \varepsilon\},\ \text{then } h(w_t)\leq \varepsilon\ \forall t\in A}{\leq} \frac{1}{|A|}\sum_{t\in A}\varepsilon = \frac{1}{|A|}|A|\varepsilon = \varepsilon,$$

So $h(\bar{w}) \leq \varepsilon$ □.

d) First, for $\eta = \frac{D}{G\sqrt{T}}$ and $\varepsilon \overset{L}{=} \frac{DG}{\sqrt{T}}$, $\frac{\eta G^2}{2} + \frac{D^2}{2\eta T} = \frac{DG^{\not 2}}{2\not G\sqrt{T}} + \frac{D^{\not 2}\sqrt{T}G}{2\not D T} = \frac{DG}{\sqrt{T}} = \varepsilon$.

By b) and above then:

$\frac{1}{T}\sum_{t\in A}(f(w_t) - f(w^*)) + \frac{1}{T}\sum_{t\in B} h(w_t) \leq \varepsilon$. Also, from definition of B it follows that $B = [T] \setminus A = \{t \in [T] \mid h(w) > \varepsilon\}$. So $h(w_t) > \varepsilon \ \forall t \in B$. ($-h(w_t) < -\varepsilon \ \forall t \in B$)

Then from above, the following holds:

$$\frac{1}{T}\sum_{t\in A}(f(w_t) - f(w^*)) \leq \varepsilon + \frac{1}{T}\sum_{t\in B}(-h(w_t)) \underset{\text{(so } \leq \text{ also holds)}}{\leq} \varepsilon + \frac{1}{T}\sum_{t\in B}(-\varepsilon) = \varepsilon - \frac{|B|}{T}\varepsilon = \frac{(T-|B|)}{T}\varepsilon.$$

Since $B = [T] \setminus A$, $T - |B| = |A|$, so: $\Leftrightarrow$

$$\frac{1}{T}\sum_{t\in A}(f(w_t) - f(w^*)) \leq \frac{|A|}{T}\varepsilon \quad / \cdot \frac{T}{|A|}$$

$\Leftrightarrow$

$$\frac{1}{|A|}\sum_{t\in A}(f(w_t) - f(w^*)) \leq \varepsilon \quad (*)$$

Finally, since f is convex:

$$f(\bar{w}) - f(w^*) = f\left(\frac{1}{|A|}\sum_{t\in A} w_t\right) - \frac{|A|}{|A|} f(w^*) \underset{\substack{\uparrow \\ f \text{ is convex}}}{\leq} \sum_{t\in A}\frac{1}{|A|} f(w_t) - \frac{\sum_{t\in A} 1}{|A|} f(w^*) =$$

$$= \frac{1}{|A|}\left(\sum_{t\in A} f(w_t) - \sum_{t\in A} f(w^*)\right) = \frac{1}{|A|}\sum_{t\in A}(f(w_t) - f(w^*)) \underset{\substack{\uparrow \\ \text{by } (*)}}{\leq} \varepsilon.$$

So: for $\eta = \frac{D}{G\sqrt{T}}$, $\varepsilon = \frac{DG}{\sqrt{T}}$, this shows that

$f(\bar{w}) - f(w^*) \leq \varepsilon$, for $\bar{w} = \frac{1}{|A|}\sum_{t\in A} w_t$ $\square$.